

10 – Node.js

Javascript on the server side

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OUTLINE

- 1) Node.js
- 2) Installation
- 3) Terminal commands
- 4) Path
- 5) Hello World!

1) NODE.JS

- **Node.js** is a **back-end Javascript runtime environment**
- It is based on the V8 Javascript Engine
- It executes Javascript code outside a web browser

Source: <https://en.wikipedia.org/wiki/Node.js>

- It can create running servers with only several lines of code

1) NODE.JS

- It can create running servers with only several lines of code

```
Create an HTTP Server Write Tests Read and Hash a File Streams Pipeline Work with Threads

1 // server.mjs
2 import { createServer } from 'node:http';
3
4 const server = createServer((req, res) => {
5   res.writeHead(200, { 'Content-Type': 'text/plain' });
6   res.end('Hello World!\n');
7 });
8
9 // starts a simple http server locally on port 3000
10 server.listen(3000, '127.0.0.1', () => {
11   console.log('Listening on 127.0.0.1:3000');
12 });
13
14 // run with `node server.mjs`

JavaScript  Copy to clipboard

Learn more what Node.js is able to offer with our Learning materials.
```

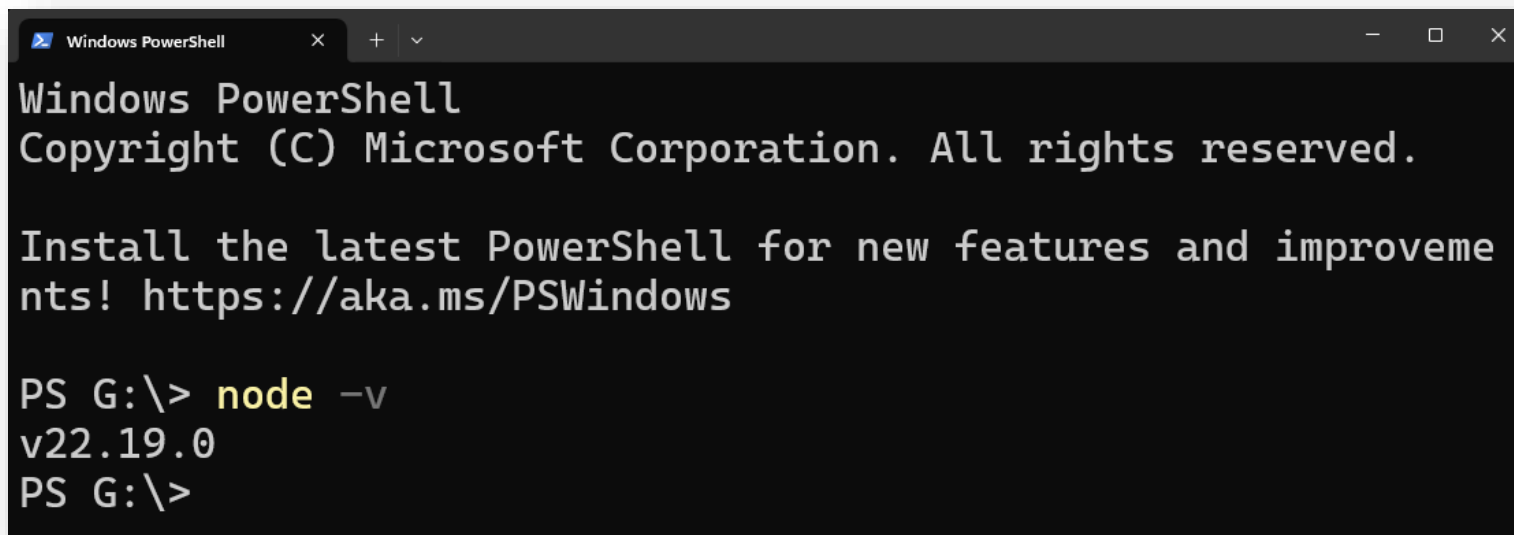
2) INSTALLATION

- Official website <https://nodejs.org>
- Two versions:
 - **LTS (Long Term Support):** The support lasts for a longer period
 - **Current:** Latest features and improvements but support limited in time



2) INSTALLATION

- On your system, to know if you have node and its version:
 - Open a terminal
 - Type the command: **node -v**
- If you obtain a **version number**, it means node is installed
- If this version number is **v22.19.0**, then you have the correct version of node installed for your project



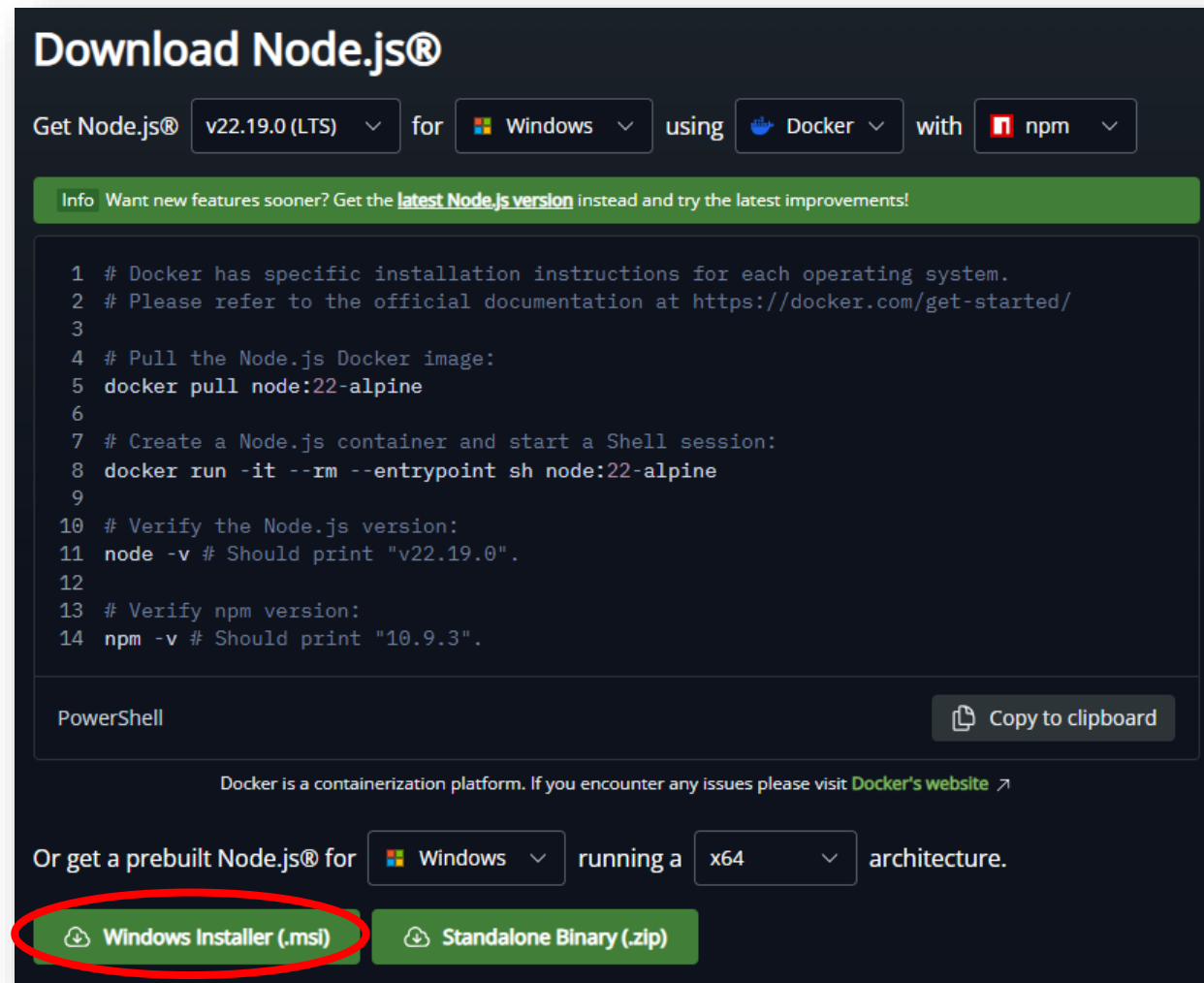
```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS G:\> node -v
v22.19.0
PS G:\>
```

2) INSTALLATION

- We will use the same version:
22.19.0 LTS (Long Term Support)
- If not installed (see the previous slide):
 - On Windows:
 - Download the installer
 - Run the installer (generally in your "Downloads" folder)
 - Click on "Next" several times...
 - Then "Finish": it's installed!



The screenshot shows the 'Download Node.js®' page. At the top, it says 'Get Node.js®' followed by a dropdown menu set to 'v22.19.0 (LTS)', 'for' a dropdown menu set to 'Windows', 'using' a dropdown menu set to 'Docker', and 'with' a dropdown menu set to 'npm'. Below this is a green banner with the text: 'Info Want new features sooner? Get the [latest Node.js version](#) instead and try the latest improvements!'. The main content area has a dark background with white text showing a list of commands for Docker installation and verification. Below the commands is a 'PowerShell' label and a 'Copy to clipboard' button. At the bottom, it says 'Or get a prebuilt Node.js® for' followed by a dropdown menu set to 'Windows', 'running a' followed by a dropdown menu set to 'x64', and 'architecture.'. At the very bottom, there are two green buttons: 'Windows Installer (.msi)' and 'Standalone Binary (.zip)'. The 'Windows Installer (.msi)' button is circled in red.

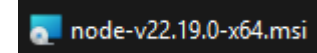
```
1 # Docker has specific installation instructions for each operating system.
2 # Please refer to the official documentation at https://docker.com/get-started/
3
4 # Pull the Node.js Docker image:
5 docker pull node:22-alpine
6
7 # Create a Node.js container and start a Shell session:
8 docker run -it --rm --entrypoint sh node:22-alpine
9
10 # Verify the Node.js version:
11 node -v # Should print "v22.19.0".
12
13 # Verify npm version:
14 npm -v # Should print "10.9.3".
```

PowerShell [Copy to clipboard](#)

Docker is a containerization platform. If you encounter any issues please visit [Docker's website](#)

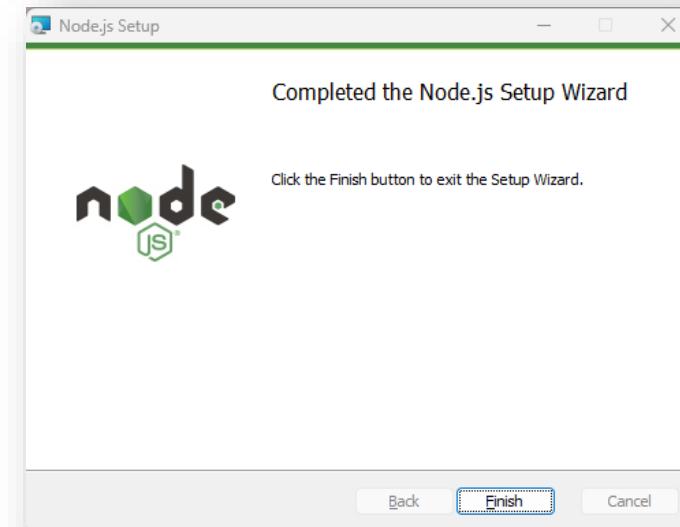
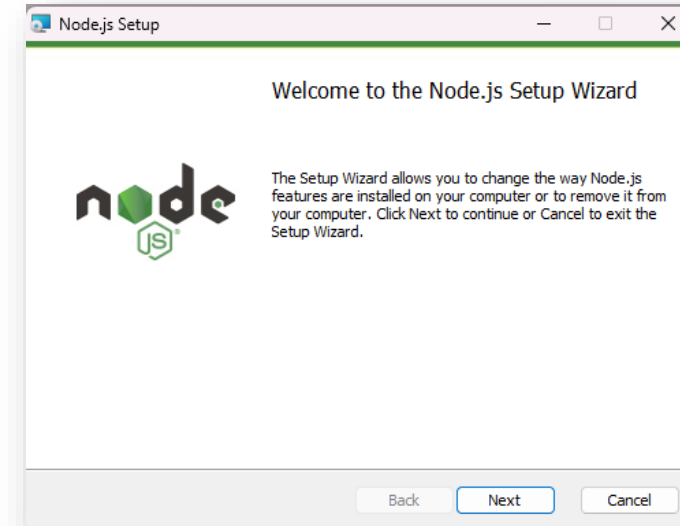
Or get a prebuilt Node.js® for [Windows](#) running a [x64](#) architecture.

[Windows Installer \(.msi\)](#) [Standalone Binary \(.zip\)](#)



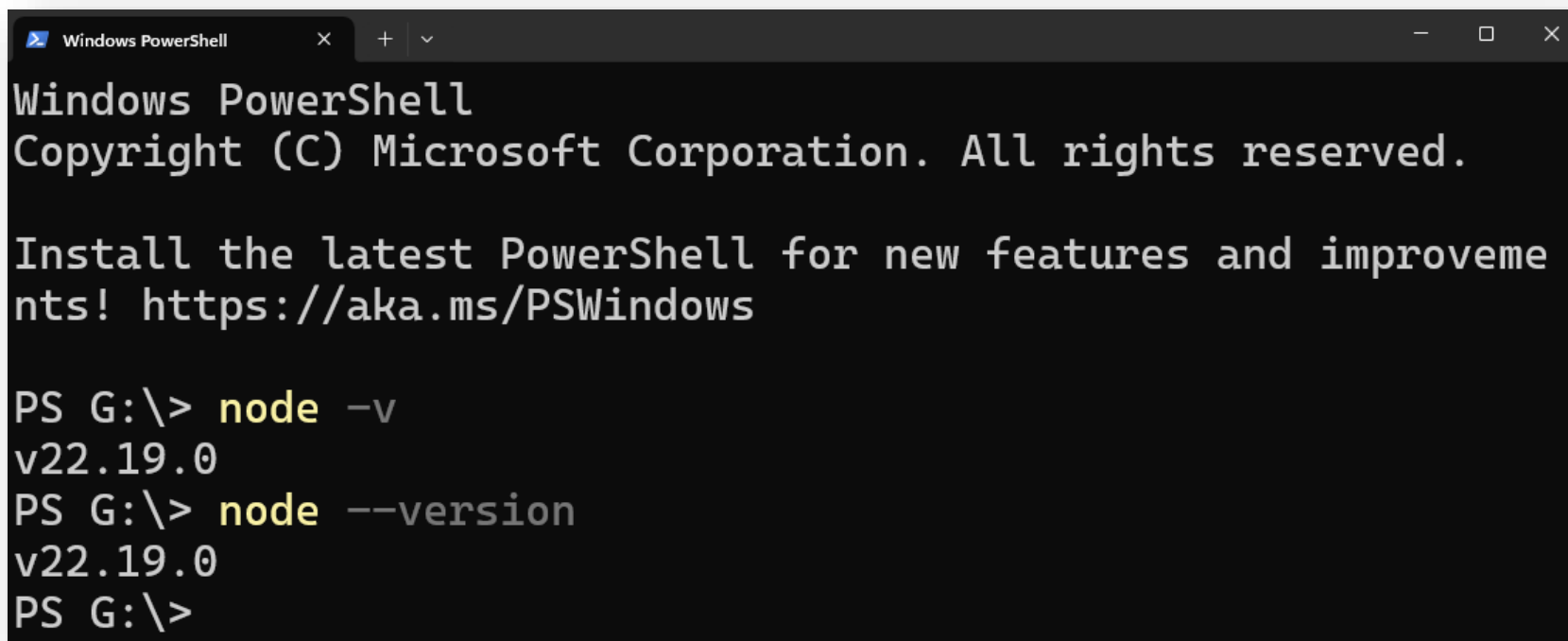
2) INSTALLATION

- We will use the same version:
22.19.0 LTS (Long Term Support)
- If not installed (see the previous slide):
 - On Windows:
 - Download the installer
 - Run the installer (generally in your "Downloads" folder)
 - Click on "Next" several times...
 - Then "Finish": it's installed!



2) INSTALLATION

- To check installation, open a terminal and type "node -v" or "node --version" at the prompt to get the version of node, if you obtain v22.19.0, it is OK!



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS G:\> node -v
v22.19.0
PS G:\> node --version
v22.19.0
PS G:\>
```

2) INSTALLATION

- You can also check the installation of npm (node package manager)
- This tool will install all the needed libraries and will verify the dependencies for you!
- Type "npm -v" to get the version of npm

```
PS G:\> npm -v  
11.2.0  
PS G:\> |
```

2) IN

- You
- Th
for you!
- Type "npm -v" to get the version of npm

```
PS G:\> npm -v  
11.2.0  
PS G:\> |
```

3) TERMINAL COMMANDS

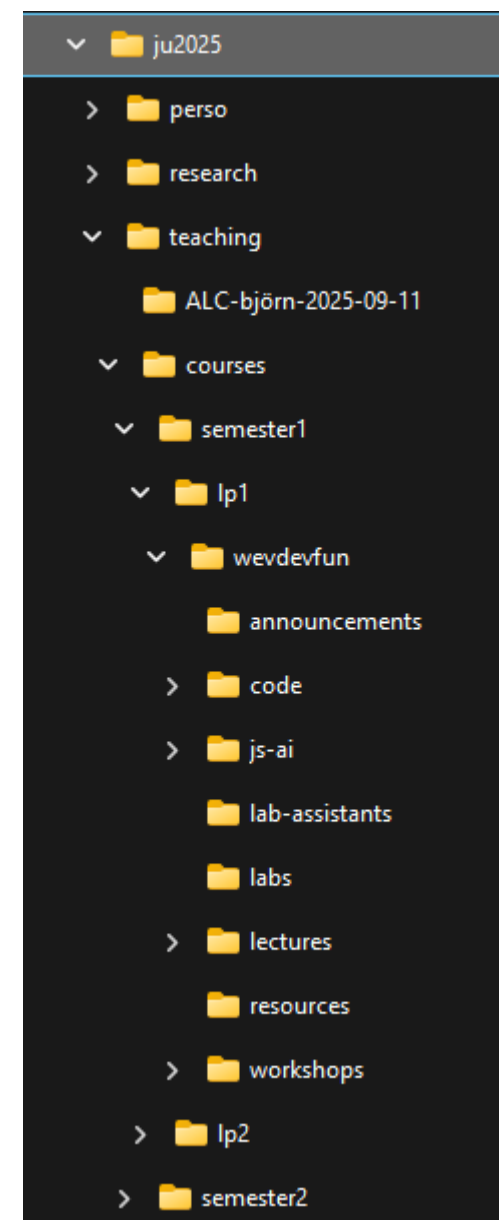
- "cd" or "pwd": tells you where you are in the file tree
- "mkdir directoryName": creates a directory in the current directory
- "cd directoryName": goes into your directoryName directory
- "cd ..": goes one level up in the file tree
- "dir" or "ls": list the files in the current directory

```
C:\Users\lanjer\Desktop\temp>cd  
C:\Users\lanjer\Desktop\temp  
  
C:\Users\lanjer\Desktop\temp>mkdir test1  
  
C:\Users\lanjer\Desktop\temp>cd test1
```

```
C:\Users\lanjer\Desktop\temp\test1>dir  
Volume in drive C is Windows  
Volume Serial Number is 2671-B1FB  
  
Directory of C:\Users\lanjer\Desktop\temp\test1  
  
2023-09-12  06:10    <DIR>          .  
2023-09-12  06:10    <DIR>          ..  
                0 File(s)                0 bytes  
                2 Dir(s)  20 341 116 928 bytes free  
  
C:\Users\lanjer\Desktop\temp\test1>cd ..  
  
C:\Users\lanjer\Desktop\temp>
```

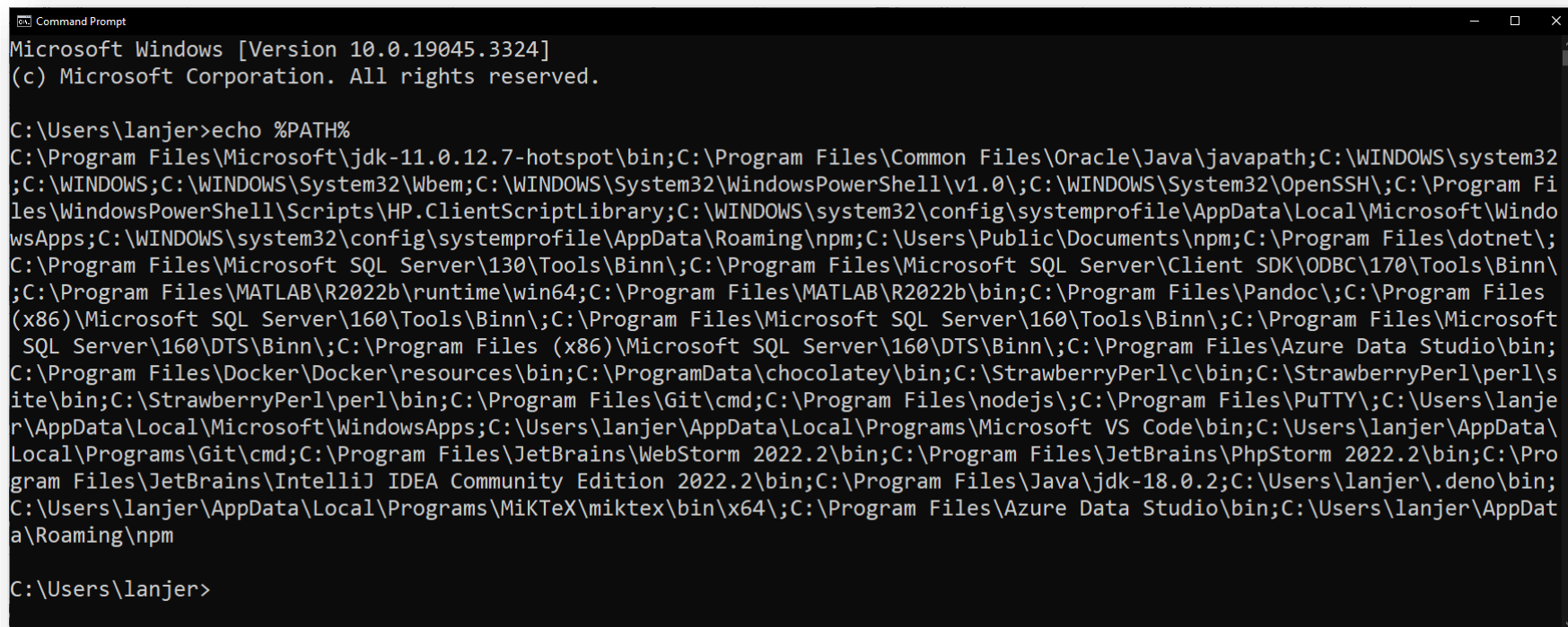
4) PATH

- On a filesystem, you have directories (folders) and files
- The filesystem is organised as a **tree** with a **root** (**/ on MacOS/Linux** and **\ on Windows**)
- In this **tree structure**, you can specify the **path** of a file or a folder.
- It is the path to follow to reach the file or folder
- There are two means to specify a path:
 - **Absolute** (you start from the root of the filesystem and you go down in folders to reach the target)
 - **Relative** (you start from the current directory and you can go up and/or down to the target)



4) PATH

- The operating system has a default **PATH** variable that will find automatically your commands
- That is why the command node is known by the system



```
Microsoft Windows [Version 10.0.19045.3324]
(c) Microsoft Corporation. All rights reserved.

C:\Users\lanjer>echo %PATH%
C:\Program Files\Microsoft\jdk-11.0.12.7-hotspot\bin;C:\Program Files\Common Files\Oracle\Java\javapath;C:\WINDOWS\system32;C:\WINDOWS;C:\WINDOWS\System32\Wbem;C:\WINDOWS\System32\WindowsPowerShell\v1.0\;C:\WINDOWS\System32\OpenSSH\;C:\Program Files\WindowsPowerShell\Scripts\HP.ClientScriptLibrary;C:\WINDOWS\system32\config\systemprofile\AppData\Local\Microsoft\WindowsApps;C:\WINDOWS\system32\config\systemprofile\AppData\Roaming\npm;C:\Users\Public\Documents\npm;C:\Program Files\dotnet\;C:\Program Files\Microsoft SQL Server\130\Tools\Binn\;C:\Program Files\Microsoft SQL Server\Client SDK\ODBC\170\Tools\Binn\;C:\Program Files\MATLAB\R2022b\runtime\win64;C:\Program Files\MATLAB\R2022b\bin;C:\Program Files\Pandoc\;C:\Program Files(x86)\Microsoft SQL Server\160\Tools\Binn\;C:\Program Files\Microsoft SQL Server\160\Tools\Binn\;C:\Program Files\Microsoft SQL Server\160\DTS\Binn\;C:\Program Files(x86)\Microsoft SQL Server\160\DTS\Binn\;C:\Program Files\Azure Data Studio\bin;C:\Program Files\Docker\Docker\resources\bin;C:\ProgramData\chocolatey\bin;C:\StrawberryPerl\c\bin;C:\StrawberryPerl\perl\site\bin;C:\StrawberryPerl\perl\bin;C:\Program Files\Git\cmd;C:\Program Files\nodejs\;C:\Program Files\PuTTY\;C:\Users\lanjer\AppData\Local\Microsoft\WindowsApps;C:\Users\lanjer\AppData\Local\Programs\Microsoft VS Code\bin;C:\Users\lanjer\AppData\Local\Programs\Git\cmd;C:\Program Files\JetBrains\WebStorm 2022.2\bin;C:\Program Files\JetBrains\PhpStorm 2022.2\bin;C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2022.2\bin;C:\Program Files\Java\jdk-18.0.2\;C:\Users\lanjer\.deno\bin;C:\Users\lanjer\AppData\Local\Programs\MiKTeX\miktex\bin\x64\;C:\Program Files\Azure Data Studio\bin;C:\Users\lanjer\AppData\Roaming\npm

C:\Users\lanjer>
```

To see your PATH:
echo %PATH%
or
echo \$PATH

4) PATH

- From the terminal, you can specify commands and path in the **absolute** or **relative** way, **you choose!**
- To run a node program, you have to run the node command on the file code
- On the prompt, you type "node" then "space" and you give the path of the file to run

```
C:\Users\lanjer\Desktop\temp\test1>dir
Volume in drive C is Windows
Volume Serial Number is 2671-B1FB

Directory of C:\Users\lanjer\Desktop\temp\test1

2023-09-12  06:28    <DIR>          .
2023-09-12  06:28    <DIR>          ..
2023-09-12  06:29                31 server-01.js
                1 File(s)                31 bytes
                2 Dir(s)  20 328 706 048 bytes free

C:\Users\lanjer\Desktop\temp\test1>node server-01.js
Hello World!
```

4) PATH

- An **absolute** path starts with the root (/ or \)
- A **relative** path doesn't start with the root (/ or \)
- With paths, you can run commands from any directory

```
C:\Users\lanjer\Desktop>node temp\test1\server-01.js  
Hello World!
```

Relative

```
C:\Users\lanjer\Desktop>node \Users\lanjer\Desktop\temp\test1\server-01.js  
Hello World!
```

Absolute

```
C:\>node Users\lanjer\Desktop\temp\test1\server-01.js  
Hello World!
```

Relative

Game: Relative
or Absolute

4) PATH

Game: Relative
or Absolute

Relative

```
C:\Users\lanjer\Desktop\temp\test1>mkdir test2  
C:\Users\lanjer\Desktop\temp\test1>cd test2  
C:\Users\lanjer\Desktop\temp\test1\test2>node ..\server-01.js  
Hello World!  
C:\Users\lanjer\Desktop\temp\test1\test2>mkdir test3  
C:\Users\lanjer\Desktop\temp\test1\test2>cd test3  
C:\Users\lanjer\Desktop\temp\test1\test2\test3>node ../../server-01.js  
Hello World!  
C:\Users\lanjer\Desktop\temp\test1\test2\test3>node \Users\lanjer\Desktop\temp\test1\server-01.js  
Hello World!
```

Relative

Relative

Relative

Relative

Relative

Relative

Absolute

5) HELLO WORLD!

- When you learn programming, the first code example is the "Hello World!" application
- It is the simple text "Hello World!" displayed on the screen of the computer
- Let's do it in Node!

5) HELLO WORLD!

- To create a Node.js application:
 - Create a folder: "mkdir my-node-app"
 - Go into it: "cd my-node-app"
 - Initialize a Node app: "npm init"
 - Answer the questions...
 - Start VSCode: "code ."

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SQL CONSOLE

• PS C:\Users\lanjer> cd .\Desktop\
• PS C:\Users\lanjer\Desktop> mkdir my-node-app

Directory: C:\Users\lanjer\Desktop

Mode                LastWriteTime         Length Name
----                -
d-----          2023-09-12    08:37             my-node-app

• PS C:\Users\lanjer\Desktop> cd .\my-node-app\
○ PS C:\Users\lanjer\Desktop\my-node-app> 
```

```
• PS C:\Users\lanjer\Desktop> cd .\my-node-app\
• PS C:\Users\lanjer\Desktop\my-node-app> npm init
This utility will walk you through creating a package.json file.
It only covers the most common items, and tries to guess sensible defaults.

See `npm help init` for definitive documentation on these fields
and exactly what they do.

Use `npm install <pkg>` afterwards to install a package and
save it as a dependency in the package.json file.

Press ^C at any time to quit.
package name: (my-node-app)
version: (1.0.0)
description: My first Node app.
entry point: (index.js) server-01.js
test command:
git repository:
keywords: First Node app, Hello World.
author: JL
license: (ISC)
```

5) HELLO WORLD!

- "npm init" creates the package.json file

```
{ } package.json > ...
1  {
2    "name": "my-node-app",
3    "version": "1.0.0",
4    "description": "My first Node app.",
5    "main": "server-01.js",
6    "scripts": {
7      "test": "echo \"Error: no test specified\" && exit 1"
8    },
9    "keywords": [
10     "First",
11     "Node",
12     "app",
13     "Hello",
14     "World."
15   ],
16   "author": "JL",
17   "license": "ISC"
18 }
19
```

```
About to write to C:\Users\lanjer\Desktop\my-node-app\package.json:

{
  "name": "my-node-app",
  "version": "1.0.0",
  "description": "My first Node app.",
  "main": "server-01.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [
    "First",
    "Node",
    "app",
    "Hello",
    "World."
  ],
  "author": "JL",
  "license": "ISC"
}

Is this OK? (yes)
PS C:\Users\lanjer\Desktop\my-node-app> █
```

5) HELLO WORLD!

```
JS server-01.js > ...
1  const http = require('http');
2
3  const port = 3000;
4
5  const server = http.createServer(function (req, res) {
6    res.statusCode = 200;
7    res.setHeader('Content-Type', 'text/plain');
8    res.end('Hello World!');
9  });
10
11 server.listen(port, function () {
12   console.log(`Server running and listening on port ${port}...`);
13 });
14
```

1) Write
the code

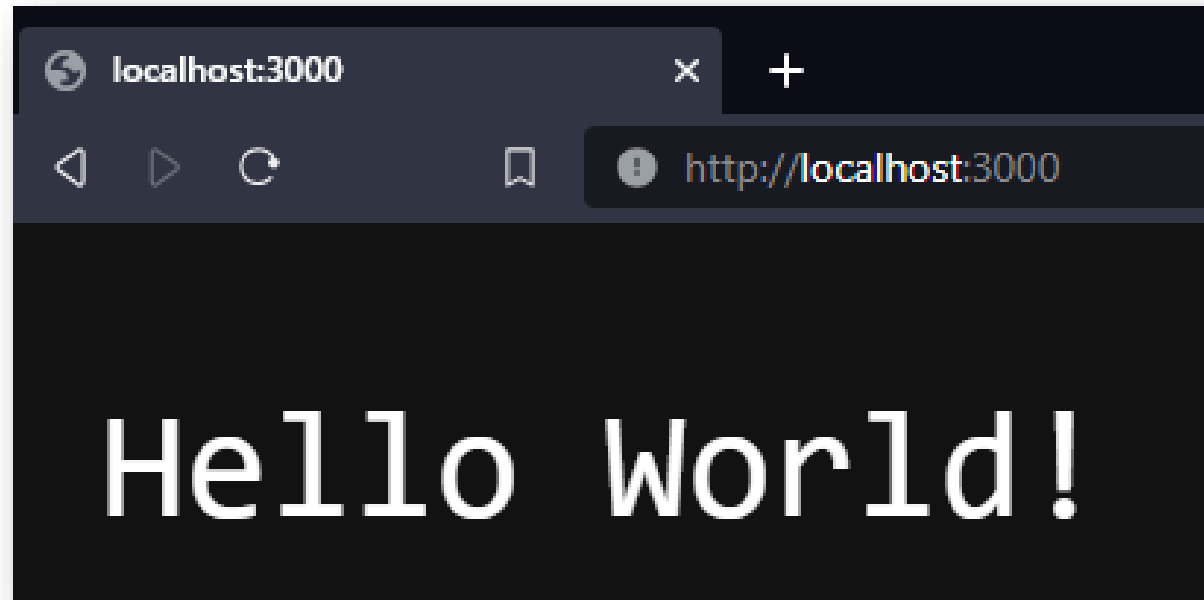
5) HELLO WORLD!

```
C:\Users\lanjer\Desktop\my-node-app>node server-01.js  
Server running and listening on port 3000...  
_
```

2) Run the code

The easiest way is to go into the directory where your server file "myserver.js" is and run "node myserver.js"

5) HELLO WORLD!



3) Test the code

5) HELLO WORLD!

You **MUST LEARN** the JS language to avoid any



5) HELLO WORLD!

"function" notation

```
1  const http = require('http');
2
3  const port = 3000;
4
5  const server = http.createServer(function (req, res) {
6    |   res.statusCode = 200;
7    |   res.setHeader('Content-Type', 'text/plain');
8    |   res.end('Hello World!');
9  });
10
11 server.listen(port, function () {
12 |   console.log(`Server running and listening on port ${port}...`);
13 | });
14
```

"=>" notation

```
1  const http = require('http');
2
3  const port = 3000;
4
5  const server = http.createServer((req, res) => {
6    |   res.statusCode = 200;
7    |   res.setHeader('Content-Type', 'text/plain');
8    |   res.end('Hello World!');
9  });
10
11 server.listen(port, () => {
12 |   console.log(`Server running and listening on port ${port}...`);
13 | });
14
```

Callback function: a function called **by Node**, not by you!

ANY QUESTIONS?

LIVE DEMONSTRATION CODING!